Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Ken Van Foeken Dairy #1

Physical address of dairy:

22338 Short RD	Hilmar	Merced	95324
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 10/01/1983

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X045-X130-X046-XXXX

B. OPERATORS

VanFoeken, Ken

Operator name: VanFoeken, Ken

Telephone no.: (209) 667-2393 Landline, (209) 658-3912 Cellular

22338 Short RD	Hilmar	CA	95324
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

VanFoeken, Ken

Legal owner name: VanFoeken, Ken

Telephone no.: (209) 667-2393 Landline, (209) 658-3912 Cellular

22338 Short RD	Hilmar	CA	95324
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	410	43	43	55	30	38
Maximum number	410	43	43	55	30	38
Average number	410	43	43	55	30	38
Avg live weight (lbs)	1,200	1,400	900	650		

Predominant milk cow breed: Jersey

Average milk production: 57 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 11,690.26 *tons per reporting period*

Total nitrogen from manure: 149,949.67 *lbs per reporting period*

After ammonia losses (30% loss applied): 104,964.77 *lbs per reporting period*

Total phosphorus from manure: 24,630.55 *lbs per reporting period*

Total potassium from manure: 75,016.97 *lbs per reporting period*

Total salt from manure: 202,936.35 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 12,660,000 *gallons*

Total nitrogen generated: 45,438.63 *lbs*

Total phosphorus generated: 7,721.73 *lbs*

Total potassium generated: 62,227.37 *lbs*

Total salt generated: 341,734.01 *lbs*

	12,660,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	12,660,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Domestic Well - Well 1	Ground water
Domestic Well - Well 2	Ground water
Irrigation Water - Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/19/2023	Corral solids	1,200.00 *ton*	Dry-weight	15.7		22,900.00	7,700.00	28,200.00		0.00

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	46,331.28	15,578.64	57,054.24	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	46,331.28	15,578.64	57,054.24	0.00

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
40	37	37	2	process wastewater	X045-X130-X033-XXXX
5	5	5	0	none	X045-X130-X049-XXXX
KA	12	12	7	process wastewater	X045-X130-X048-XXXX
KA20	18	18	2	process wastewater	X045-X130-X031-XXXX
KVF	10	10	2	process wastewater	X045-X130-X046-XXXX
LF	17	17	2	process wastewater	X045-X130-X045-XXXX
Totals for areas that were used for application	94	94	15		
Totals for areas that were not used for application	5	5	0		
Land application area totals	99	99	15		

B. CROPS AND HARVESTS

40

Field name: 40

11/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 37 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	334.60 *ton*	Dry-weight		52.7	18,100.00	3,800.00	24,900.00		10.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	9.04	154.84	32.51	213.02	932.48

40

05/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 37 Plant date: 05/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/11/2023	962.10 *ton*	Dry-weight		67.1	15,500.00	3,200.00	17,500.00		6.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.00	265.20	54.75	299.42	1,026.59

KA

Field name: KA

10/30/2021: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 12 Plant date: 10/30/2021

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	13.90 *ton*	Dry-weight		12.4	27,700.00	3,000.00	22,900.00		9.40
05/16/2023	11.90 *ton*	Dry-weight		12.4	27,700.00	3,000.00	22,900.00		9.40
06/13/2023	12.30 *ton*	Dry-weight		12.4	27,700.00	3,000.00	22,900.00		9.40
07/11/2023	12.20 *ton*	Dry-weight		12.4	27,700.00	3,000.00	22,900.00		9.40
08/08/2023	12.40 *ton*	Dry-weight		12.4	27,700.00	3,000.00	22,900.00		9.40
09/25/2023	11.45 *ton*	Dry-weight		12.4	27,700.00	3,000.00	22,900.00		9.40
10/28/2023	10.80 *ton*	Dry-weight		12.4	27,700.00	3,000.00	22,900.00		9.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.08	343.55	37.21	284.02	1,165.85

KA20

Field name: KA20

KA20

11/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 18 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	165.30 *ton*	Dry-weight		48.0	16,900.00	4,100.00	31,100.00		8.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	9.18	161.41	39.16	297.03	811.81

05/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 18 Plant date: 05/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/11/2023	481.50 *ton*	Dry-weight		71.6	15,300.00	2,800.00	16,200.00		6.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.75	232.47	42.54	246.14	957.22

KVF

Field name: KVF

11/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 10 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	93.60 *ton*	Dry-weight		48.0	16,900.00	4,100.00	31,100.00		8.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	9.36	164.51	39.91	302.74	827.42

KVF

05/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 10 Plant date: 05/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/11/2023	264.30 *ton*	Dry-weight		71.6	15,300.00	2,800.00	16,200.00		6.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.43	229.69	42.03	243.20	945.77

LF

Field name: LF

11/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 17 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	136.00 *ton*	Dry-weight		34.6	20,600.00	5,200.00	37,800.00		12.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	8.00	215.56	54.41	395.54	1,255.68

05/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 17 Plant date: 05/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/11/2023	408.00 *ton*	Dry-weight		60.5	14,800.00	3,200.00	14,800.00		5.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	24.00	280.61	60.67	280.61	1,118.64

NUTRIENT BUDGET

A. LAND APPLICATIONS

40 - 11/05/2022: Oats, silage-soft dough

Field name: 40

Crop: Oats, silage-soft dough

Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	98.61	16.81	134.73	744.45	1,012,500.00 *gal*
Irrigation Water - Canal	Surface water	3.70	0.00	0.00	91.82	5,219,625.00 *gal*
Application event totals		102.30	16.81	134.73	836.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	97.88	16.68	133.73	738.94	1,005,000.00 *gal*
Irrigation Water - Canal	Surface water	3.60	0.00	0.00	89.45	5,084,925.00 *gal*
Application event totals		101.48	16.68	133.73	828.39	

40 - 05/26/2023: Corn, silage

Field name: 40

Crop: Corn, silage

Plant date: 05/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	3.70	0.00	0.00	91.82	5,219,625.00 *gal*
Application event totals		3.70	0.00	0.00	91.82	

40 - 05/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	109.56	18.67	149.70	827.17	1,125,000.00 *gal*
Irrigation Water - Canal	Surface water	3.60	0.00	0.00	89.45	5,084,925.00 *gal*
Application event totals		113.16	18.67	149.70	916.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	102.99	17.55	140.72	777.54	1,057,500.00 *gal*
Irrigation Water - Canal	Surface water	3.53	0.00	0.00	87.68	4,983,900.00 *gal*
Application event totals		106.52	17.55	140.72	865.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	3.51	0.00	0.00	87.09	4,950,225.00 *gal*
Application event totals		3.51	0.00	0.00	87.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	95.68	16.31	130.74	722.40	982,500.00 *gal*
Irrigation Water - Canal	Surface water	3.55	0.00	0.00	88.27	5,017,575.00 *gal*
Application event totals		99.24	16.31	130.74	810.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	3.58	0.00	0.00	88.86	5,051,250.00 *gal*
Application event totals		3.58	0.00	0.00	88.86	

40 - 05/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	3.48	0.00	0.00	86.49	4,916,550.00 *gal*
Application event totals		3.48	0.00	0.00	86.49	

08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	3.31	0.00	0.00	82.35	4,680,825.00 *gal*
Application event totals		3.31	0.00	0.00	82.35	

KA - 10/30/2021: Alfalfa, hay

Field name: KA

Crop: Alfalfa, hay

Plant date: 10/30/2021

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	6.10	0.00	0.00	151.61	2,795,025.00 *gal*
Application event totals		6.10	0.00	0.00	151.61	

05/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	110.35	18.81	150.78	833.14	367,500.00 *gal*
Irrigation Water - Canal	Surface water	5.51	0.00	0.00	137.00	2,525,625.00 *gal*
Application event totals		115.87	18.81	150.78	970.14	

06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	4.93	0.00	0.00	122.38	2,256,225.00 *gal*
Application event totals		4.93	0.00	0.00	122.38	

KA - 10/30/2021: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	101.34	17.27	138.47	765.13	337,500.00 *gal*
Irrigation Water - Canal	Surface water	4.63	0.00	0.00	115.08	2,121,525.00 *gal*
Application event totals		105.98	17.27	138.47	880.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	4.93	0.00	0.00	122.38	2,256,225.00 *gal*
Application event totals		4.93	0.00	0.00	122.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	99.09	16.89	135.40	748.13	330,000.00 *gal*
Irrigation Water - Canal	Surface water	4.49	0.00	0.00	111.42	2,054,175.00 *gal*
Application event totals		103.58	16.89	135.40	859.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	4.78	0.00	0.00	118.73	2,188,875.00 *gal*
Application event totals		4.78	0.00	0.00	118.73	

KA20 - 11/05/2022: Oats, silage-soft dough

Field name: KA20

Crop: Oats, silage-soft dough

Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

KA20 - 11/05/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	103.60	17.66	141.55	782.14	517,500.00 *gal*
Irrigation Water - Canal	Surface water	3.58	0.00	0.00	88.90	2,458,275.00 *gal*
Application event totals		107.18	17.66	141.55	871.03	

03/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	100.59	17.15	137.45	759.46	502,500.00 *gal*
Irrigation Water - Canal	Surface water	4.26	0.00	0.00	105.94	2,929,725.00 *gal*
Application event totals		104.86	17.15	137.45	865.41	

KA20 - 05/26/2023: Corn, silage

Field name: KA20

Crop: Corn, silage

Plant date: 05/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	3.43	0.00	0.00	85.24	2,357,250.00 *gal*
Application event totals		3.43	0.00	0.00	85.24	

06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	96.09	16.38	131.29	725.46	480,000.00 *gal*
Irrigation Water - Canal	Surface water	3.97	0.00	0.00	98.64	2,727,675.00 *gal*
Application event totals		100.06	16.38	131.29	824.10	

KA20 - 05/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	108.10	18.43	147.71	816.14	540,000.00 *gal*
Irrigation Water - Canal	Surface water	3.28	0.00	0.00	81.59	2,256,225.00 *gal*
Application event totals		111.39	18.43	147.71	897.73	

07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	3.28	0.00	0.00	81.59	2,256,225.00 *gal*
Application event totals		3.28	0.00	0.00	81.59	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	83.08	12.87	121.37	554.04	465,000.00 *gal*
Irrigation Water - Canal	Surface water	3.68	0.00	0.00	91.33	2,525,625.00 *gal*
Application event totals		86.76	12.87	121.37	645.37	

08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	3.28	0.00	0.00	81.59	2,256,225.00 *gal*
Application event totals		3.28	0.00	0.00	81.59	

08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	3.09	0.00	0.00	76.72	2,121,525.00 *gal*
Application event totals		3.09	0.00	0.00	76.72	

08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	4.02	0.00	0.00	99.86	2,761,350.00 *gal*
Application event totals		4.02	0.00	0.00	99.86	

KVF - 11/05/2022: Oats, silage-soft dough

Field name: KVF

Crop: Oats, silage-soft dough

Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	121.61	20.73	166.17	918.16	337,500.00 *gal*
Irrigation Water - Canal	Surface water	4.32	0.00	0.00	107.41	1,650,075.00 *gal*
Application event totals		125.94	20.73	166.17	1,025.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	83.78	14.28	114.47	632.51	232,500.00 *gal*
Irrigation Water - Canal	Surface water	3.35	0.00	0.00	83.29	1,279,650.00 *gal*
Application event totals		87.13	14.28	114.47	715.80	

KVF - 05/26/2023: Corn, silage

Field name: KVF

Crop: Corn, silage

Plant date: 05/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	4.85	0.00	0.00	120.56	1,852,125.00 *gal*
Application event totals		4.85	0.00	0.00	120.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	113.51	19.35	155.09	856.95	315,000.00 *gal*
Irrigation Water - Canal	Surface water	3.62	0.00	0.00	89.87	1,380,675.00 *gal*
Application event totals		117.12	19.35	155.09	946.82	

KVF - 05/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	91.89	15.66	125.55	693.72	255,000.00 *gal*
Irrigation Water - Canal	Surface water	3.97	0.00	0.00	98.64	1,515,375.00 *gal*
Application event totals		95.86	15.66	125.55	792.36	

07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	3.35	0.00	0.00	83.29	1,279,650.00 *gal*
Application event totals		3.35	0.00	0.00	83.29	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	78.37	13.36	107.09	591.70	217,500.00 *gal*
Irrigation Water - Canal	Surface water	3.79	0.00	0.00	94.25	1,448,025.00 *gal*
Application event totals		82.17	13.36	107.09	685.96	

08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	3.62	0.00	0.00	89.87	1,380,675.00 *gal*
Application event totals		3.62	0.00	0.00	89.87	

08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	3.44	0.00	0.00	85.49	1,313,325.00 *gal*
Application event totals		3.44	0.00	0.00	85.49	

08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	3.35	0.00	0.00	83.29	1,279,650.00 *gal*
Application event totals		3.35	0.00	0.00	83.29	

LF - 11/05/2022: Oats, silage-soft dough

Field name: LF

Crop: Oats, silage-soft dough

Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	106.51	18.15	145.53	804.14	502,500.00 *gal*
Irrigation Water - Canal	Surface water	2.91	0.00	0.00	72.21	1,885,800.00 *gal*
Application event totals		109.42	18.15	145.53	876.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	112.87	19.24	154.22	852.15	532,500.00 *gal*
Irrigation Water - Canal	Surface water	3.22	0.00	0.00	79.94	2,087,850.00 *gal*
Application event totals		116.09	19.24	154.22	932.09	

LF - 05/26/2023: Corn, silage

Field name: LF

Crop: Corn, silage

Plant date: 05/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	3.27	0.00	0.00	81.23	2,121,525.00 *gal*
Application event totals		3.27	0.00	0.00	81.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	120.82	20.59	165.08	912.16	570,000.00 *gal*
Irrigation Water - Canal	Surface water	3.32	0.00	0.00	82.52	2,155,200.00 *gal*
Application event totals		124.14	20.59	165.08	994.68	

LF - 05/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	106.51	18.15	145.53	804.14	502,500.00 *gal*
Irrigation Water - Canal	Surface water	3.48	0.00	0.00	86.39	2,256,225.00 *gal*
Application event totals		109.99	18.15	145.53	890.53	

07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	3.01	0.00	0.00	74.78	1,953,150.00 *gal*
Application event totals		3.01	0.00	0.00	74.78	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	100.15	17.07	136.85	756.13	472,500.00 *gal*
Irrigation Water - Canal	Surface water	2.85	0.00	0.00	70.92	1,852,125.00 *gal*
Application event totals		103.01	17.07	136.85	827.05	

08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	3.27	0.00	0.00	81.23	2,121,525.00 *gal*
Application event totals		3.27	0.00	0.00	81.23	

08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	2.60	0.00	0.00	64.47	1,683,750.00 *gal*
Application event totals		2.60	0.00	0.00	64.47	

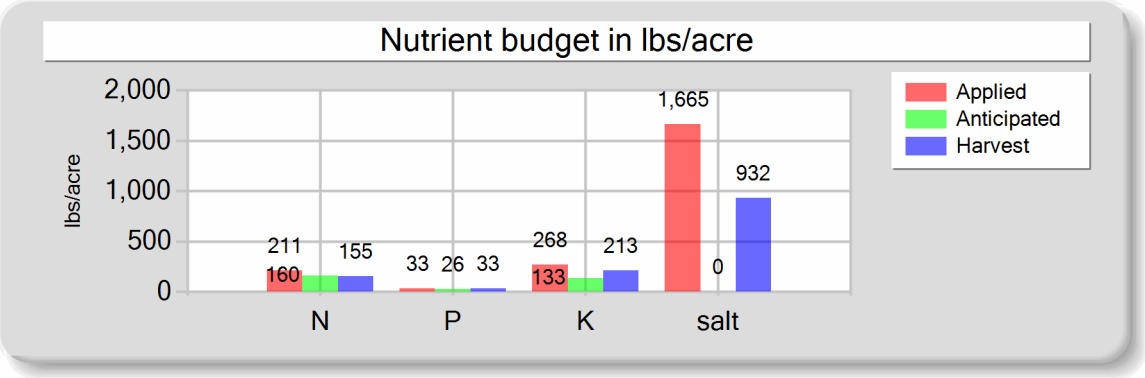
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	3.17	0.00	0.00	78.65	2,054,175.00 *gal*
Application event totals		3.17	0.00	0.00	78.65	

B. NUTRIENT BUDGET

40 - 11/05/2022: Oats, silage-soft dough

Field name: 40 Crop: Oats, silage-soft dough Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	196.48	33.49	268.47	1,483.39
Fresh water	7.30	0.00	0.00	181.28
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	210.78	33.49	268.47	1,664.67
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	154.84	32.51	213.02	932.48
Nutrient balance	55.93	0.98	55.45	732.19
Applied to removed ratio	1.36	1.03	1.26	1.79

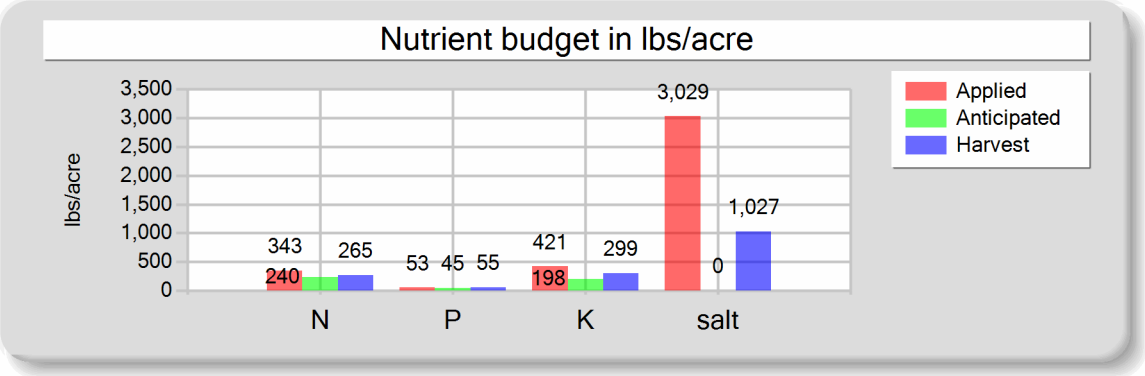
Fresh water applied	
10,304,550.00	*gallons*
379.48	*acre-inches*
10.26	*inches/acre*

Process wastewater applied	
2,017,500.00	*gallons*
74.30	*acre-inches*
2.01	*inches/acre*

Total harvests for the crop	
1	*harvests*

40 - 05/26/2023: Corn, silage

Field name: 40 Crop: Corn, silage Plant date: 05/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	308.23	52.54	421.16	2,327.10
Fresh water	28.26	0.00	0.00	702.01
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	343.49	52.54	421.16	3,029.12
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	265.20	54.75	299.42	1,026.59
Nutrient balance	78.29	-2.21	121.74	2,002.53
Applied to removed ratio	1.30	0.96	1.41	2.95

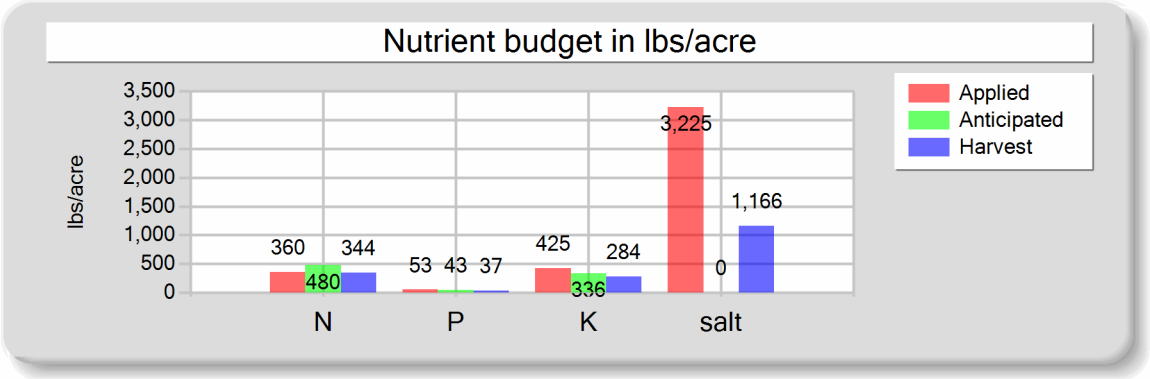
Fresh water applied	
39,904,875.00	*gallons*
1,469.56	*acre-inches*
39.72	*inches/acre*

Process wastewater applied	
3,165,000.00	*gallons*
116.56	*acre-inches*
3.15	*inches/acre*

Total harvests for the crop	
1	*harvests*

KA - 10/30/2021: Alfalfa, hay

Field name: KA | Crop: Alfalfa, hay | Plant date: 10/30/2021



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	310.79	52.97	424.66	2,346.41
Fresh water	35.37	0.00	0.00	878.60
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	360.16	52.97	424.66	3,225.01
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	343.55	37.21	284.02	1,165.85
Nutrient balance	16.61	15.77	140.63	2,059.15
Applied to removed ratio	1.05	1.42	1.50	2.77

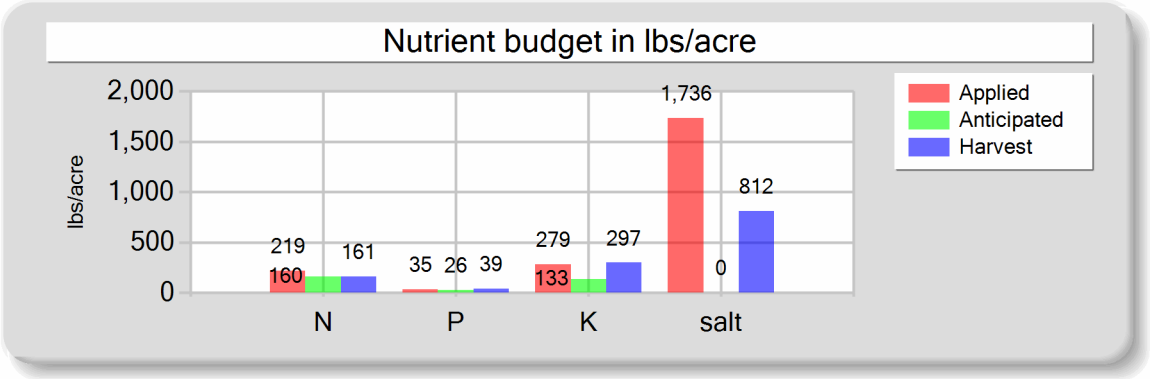
Fresh water applied	
16,197,675.00	*gallons*
596.51	*acre-inches*
49.71	*inches/acre*

Process wastewater applied	
1,035,000.00	*gallons*
38.12	*acre-inches*
3.18	*inches/acre*

Total harvests for the crop	
7	*harvests*

KA20 - 11/05/2022: Oats, silage-soft dough

Field name: KA20 Crop: Oats, silage-soft dough Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	204.19	34.80	279.00	1,541.60
Fresh water	7.84	0.00	0.00	194.84
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	219.03	34.80	279.00	1,736.44
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	161.41	39.16	297.03	811.81
Nutrient balance	57.63	-4.35	-18.02	924.63
Applied to removed ratio	1.36	0.89	0.94	2.14

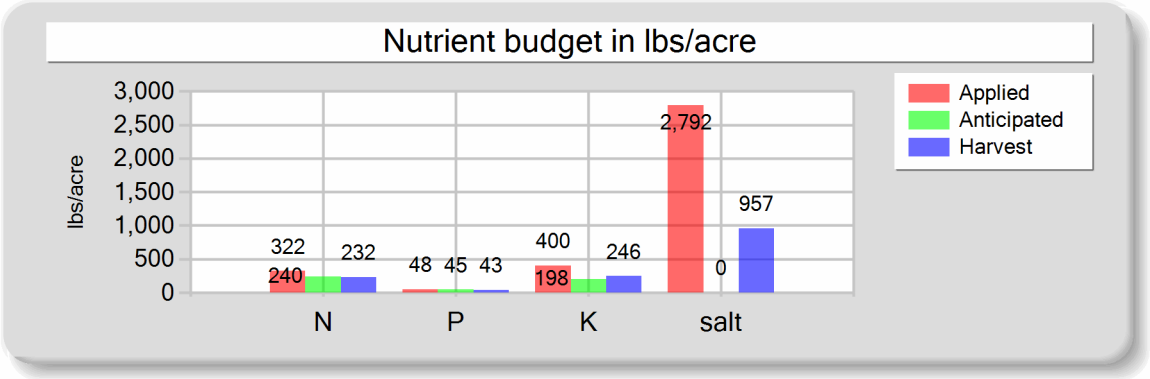
Fresh water applied	
5,388,000.00	*gallons*
198.42	*acre-inches*
11.02	*inches/acre*

Process wastewater applied	
1,020,000.00	*gallons*
37.56	*acre-inches*
2.09	*inches/acre*

Total harvests for the crop	
1	*harvests*

KA20 - 05/26/2023: Corn, silage

Field name: KA20 Crop: Corn, silage Plant date: 05/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	287.28	47.67	400.37	2,095.64
Fresh water	28.04	0.00	0.00	696.55
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	322.32	47.67	400.37	2,792.19
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	232.47	42.54	246.14	957.22
Nutrient balance	89.85	5.13	154.23	1,834.97
Applied to removed ratio	1.39	1.12	1.63	2.92

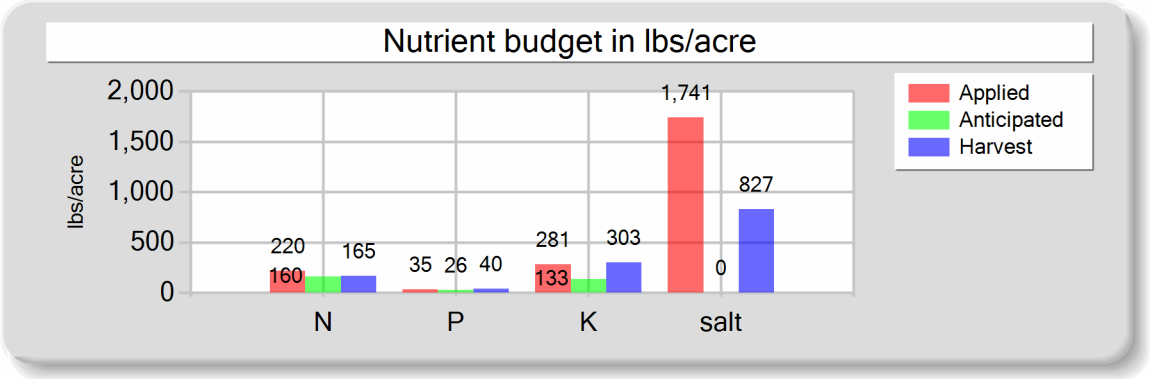
Fresh water applied	
19,262,100.00	*gallons*
709.36	*acre-inches*
39.41	*inches/acre*

Process wastewater applied	
1,485,000.00	*gallons*
54.69	*acre-inches*
3.04	*inches/acre*

Total harvests for the crop	
1	*harvests*

KVF - 11/05/2022: Oats, silage-soft dough

Field name: KVF | Crop: Oats, silage-soft dough | Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	205.39	35.01	280.64	1,550.67
Fresh water	7.68	0.00	0.00	190.70
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	220.07	35.01	280.64	1,741.37
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	164.51	39.91	302.74	827.42
Nutrient balance	55.56	-4.90	-22.10	913.94
Applied to removed ratio	1.34	0.88	0.93	2.10

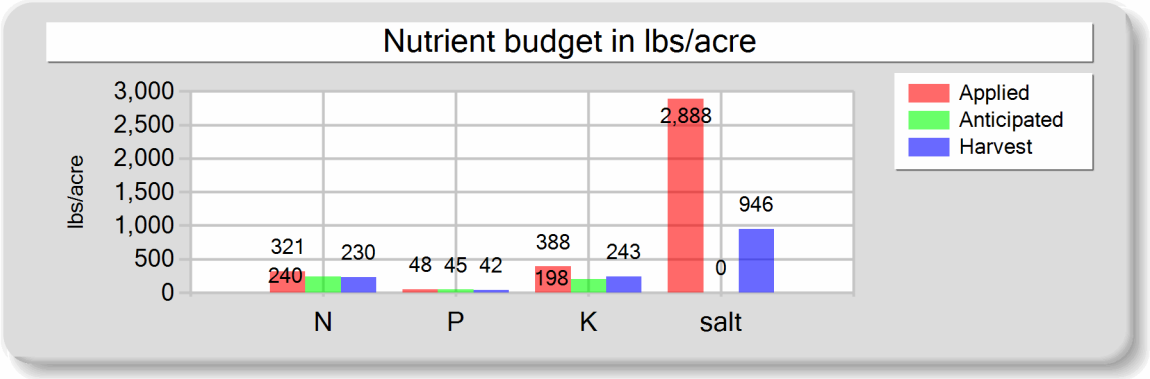
Fresh water applied	
2,929,725.00	*gallons*
107.89	*acre-inches*
10.79	*inches/acre*

Process wastewater applied	
570,000.00	*gallons*
20.99	*acre-inches*
2.10	*inches/acre*

Total harvests for the crop	
1	*harvests*

KVF - 05/26/2023: Corn, silage

Field name: KVF | Crop: Corn, silage | Plant date: 05/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	283.77	48.37	387.73	2,142.37
Fresh water	30.00	0.00	0.00	745.26
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	320.77	48.37	387.73	2,887.63
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	229.69	42.03	243.20	945.77
Nutrient balance	91.08	6.33	144.53	1,941.86
Applied to removed ratio	1.40	1.15	1.59	3.05

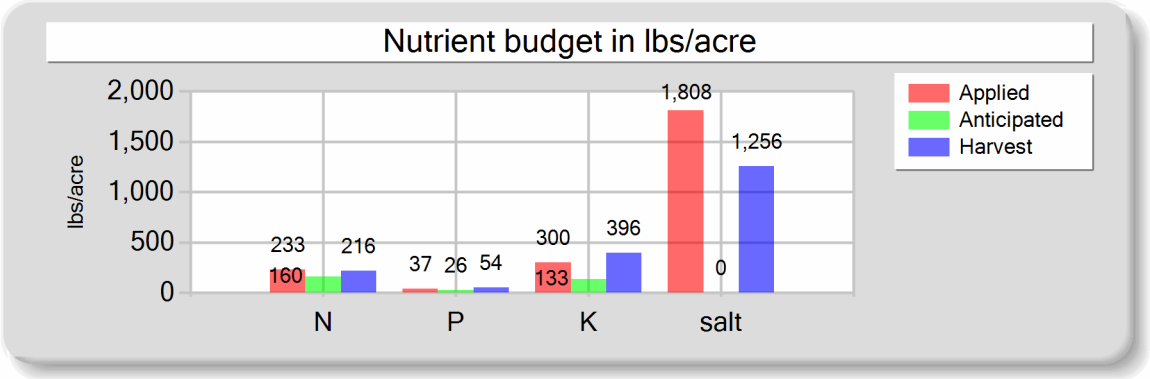
Fresh water applied	
11,449,500.00	*gallons*
421.65	*acre-inches*
42.16	*inches/acre*

Process wastewater applied	
787,500.00	*gallons*
29.00	*acre-inches*
2.90	*inches/acre*

Total harvests for the crop	
1	*harvests*

LF - 11/05/2022: Oats, silage-soft dough

Field name: LF | Crop: Oats, silage-soft dough | Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	219.38	37.39	299.76	1,656.29
Fresh water	6.12	0.00	0.00	152.15
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	232.51	37.39	299.76	1,808.43
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	215.56	54.41	395.54	1,255.68
Nutrient balance	16.95	-17.02	-95.78	552.75
Applied to removed ratio	1.08	0.69	0.76	1.44

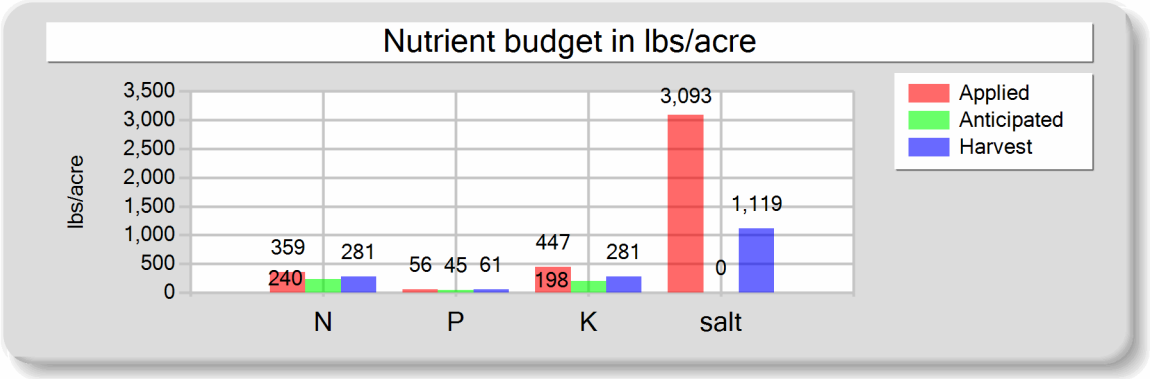
Fresh water applied	
3,973,650.00	*gallons*
146.34	*acre-inches*
8.61	*inches/acre*

Process wastewater applied	
1,035,000.00	*gallons*
38.12	*acre-inches*
2.24	*inches/acre*

Total harvests for the crop	
1	*harvests*

LF - 05/26/2023: Corn, silage

Field name: LF | Crop: Corn, silage | Plant date: 05/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	327.48	55.82	447.46	2,472.43
Fresh water	24.97	0.00	0.00	620.19
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	359.45	55.82	447.46	3,092.62
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	280.61	60.67	280.61	1,118.64
Nutrient balance	78.84	-4.85	166.86	1,973.98
Applied to removed ratio	1.28	0.92	1.59	2.76

Fresh water applied	
16,197,675.00	*gallons*
596.51	*acre-inches*
35.09	*inches/acre*

Process wastewater applied	
1,545,000.00	*gallons*
56.90	*acre-inches*
3.35	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Solid Manure

Sample and source description: Solid Manure
Sample date: 05/22/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 15.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,900.00	7,700.00	28,200.00							
DL	100.00	100.00	100.00							

Solid Manure

Sample and source description: Solid Manure
Sample date: 11/27/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 34.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,300.00	6,800.00	20,300.00							
DL	100.00	100.00	100.00							

B. PROCESS WASTEWATER ANALYSES

Lagoon Water

Sample and source description: Lagoon Water
Sample date: 03/29/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	431.00	299.00	0.00	0.80	73.60	590.00								6,660.00	3,260
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 06/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	385.00	274.00	0.00	0.40	59.70	563.00								6,150.00	2,570
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 09/25/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	270.00	123.00	0.00	13.20	69.90	240.00								2,790.00	1,670
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 11/29/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	344.00	298.00	0.00	0.10	52.20	817.00								8,120.00	4,450
DL	1.00	0.50	1.00	0.10	0.10	0.50								10.00	10

C. FRESH WATER ANALYSES

Domestic Well - Well 1

Domestic Well - Well 1

Well 1

Sample description: Well 1

Sample date: 11/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			54.80								1,150.00	
DL			0.10								10.00	

Domestic Well - Well 2

Well 2

Sample description: Well 2

Sample date: 11/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			65.50								1,330.00	
DL			0.10								10.00	

Irrigation Water - Canal

Canal

Sample description: Canal

Sample date: 09/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	3.14		2.80								117.00	78
DL	1.00		0.10								10.00	10

D. SOIL ANALYSES

40

40

Sample and source description: 40

Sample date: 12/06/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			55.40				
DL			2.00				

KA20

WST 20

Sample and source description: WST 20

Sample date: 12/12/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			47.70				
DL			2.00				

LF

C 20

Sample and source description: C 20

Sample date: 12/06/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			37.10				
DL			2.00				

E. PLANT TISSUE ANALYSES

40 - 11/05/2022: Oats, silage-soft dough

40 - 11/05/2022: Oats, silage-soft dough

40

Sample and source description: 40

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 52.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,100.00	3,800.00	24,900.00		10.90
DL	100.00	100.00	100.00		0.01

40 - 05/26/2023: Corn, silage

40

Sample and source description: 40

Sample date: 09/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,500.00	3,200.00	17,500.00		6.00
DL	100.00	100.00	100.00		0.01

KA - 10/30/2021: Alfalfa, hay

KA

Sample and source description: KA

Sample date: 04/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 16.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	44,700.00	4,400.00	25,600.00		12.90
DL	100.00	100.00	100.00		0.01

KA - 10/30/2021: Alfalfa, hay

KA

Sample and source description: KA

Sample date: 09/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 12.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,700.00	3,000.00	22,900.00		9.40
DL	100.00	100.00	100.00		0.01

KA20 - 11/05/2022: Oats, silage-soft dough

KA20

Sample and source description: KA20

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 48.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,900.00	4,100.00	31,100.00		8.50
DL	100.00	100.00	100.00		0.01

KA20 - 05/26/2023: Corn, silage

KA20

Sample and source description: KA20

Sample date: 09/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,300.00	2,800.00	16,200.00		6.30
DL	100.00	100.00	100.00		0.01

KVF - 11/05/2022: Oats, silage-soft dough

KVF

Sample and source description: KVF

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 48.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,900.00	4,100.00	31,100.00		8.50
DL	100.00	100.00	100.00		0.01

KVF - 05/26/2023: Corn, silage

KVF

Sample and source description: KVF

Sample date: 09/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,300.00	2,800.00	16,200.00		6.30
DL	100.00	100.00	100.00		0.01

LF - 11/05/2022: Oats, silage-soft dough

LF

Sample and source description: LF

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 34.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,600.00	5,200.00	37,800.00		12.00
DL	100.00	100.00	100.00		0.01

LF - 05/26/2023: Corn, silage

LF

Sample and source description: LF

Sample date: 09/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,800.00	3,200.00	14,800.00		5.90
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

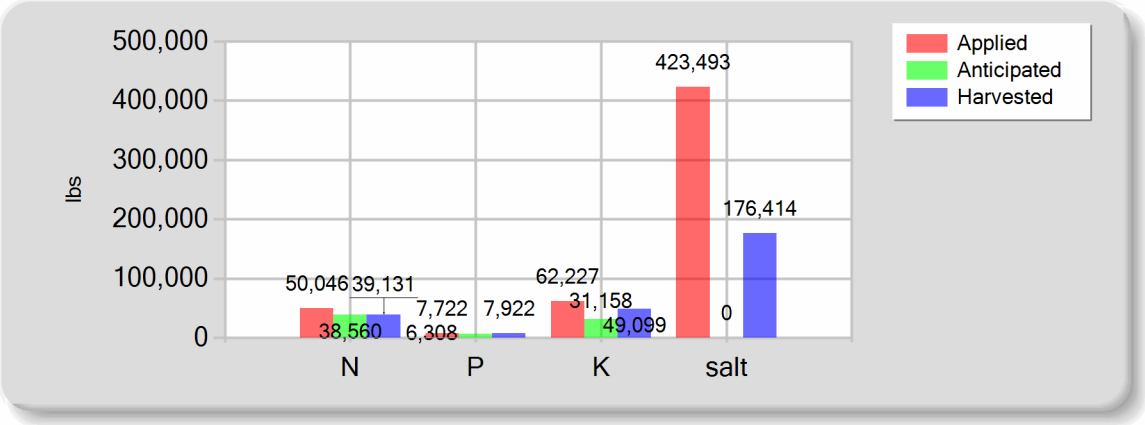
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

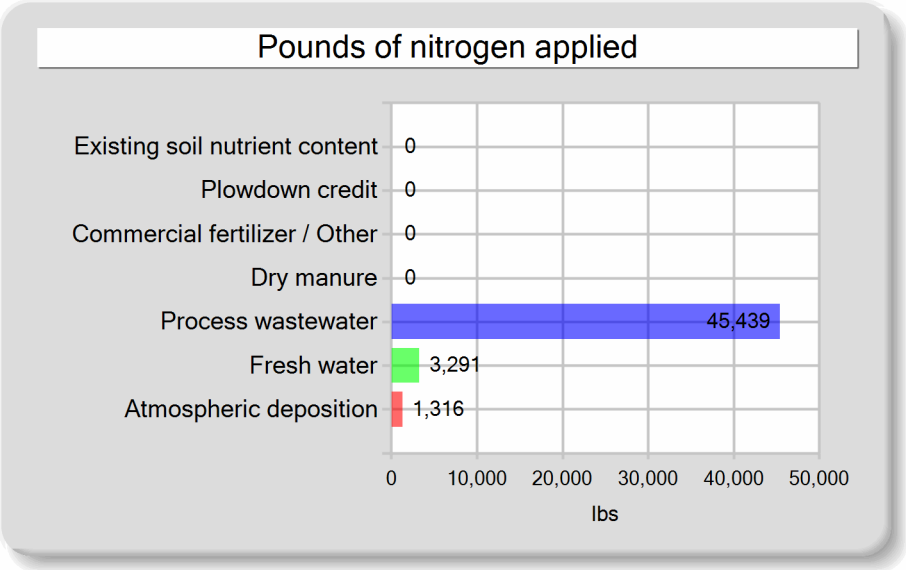
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

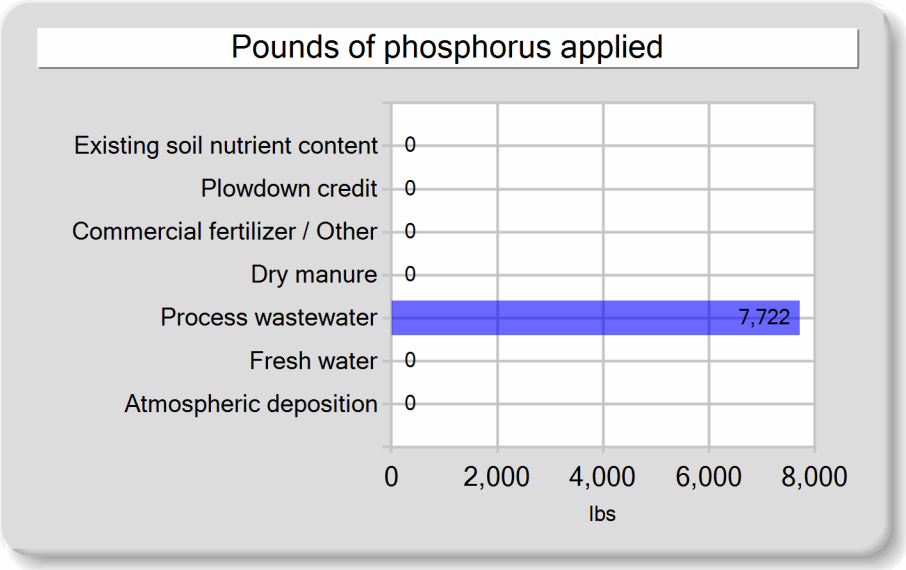
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	45,438.63	7,721.73	62,227.37	341,734.01
Fresh water	3,291.34	0.00	0.00	81,759.34
Atmospheric deposition	1,316.00	0.00	0.00	0.00
Total nutrients applied	50,045.96	7,721.73	62,227.37	423,493.35
Anticipated crop nutrient removal	38,560.00	6,307.60	31,157.60	0.00
Actual crop nutrient removal	39,130.89	7,921.63	49,099.40	176,413.81
Nutrient balance	10,915.07	-199.89	13,127.97	247,079.54
Applied to removed ratio	1.28	0.97	1.27	2.40

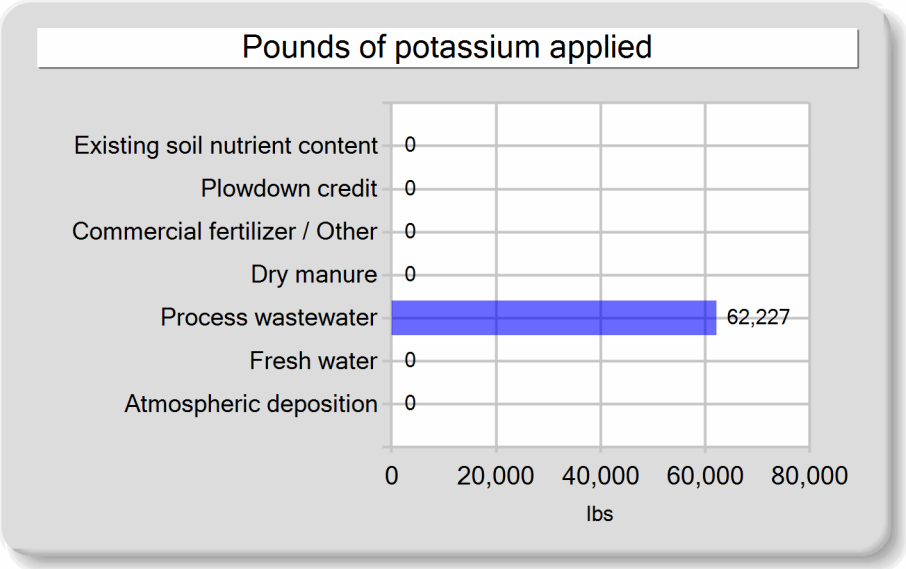
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

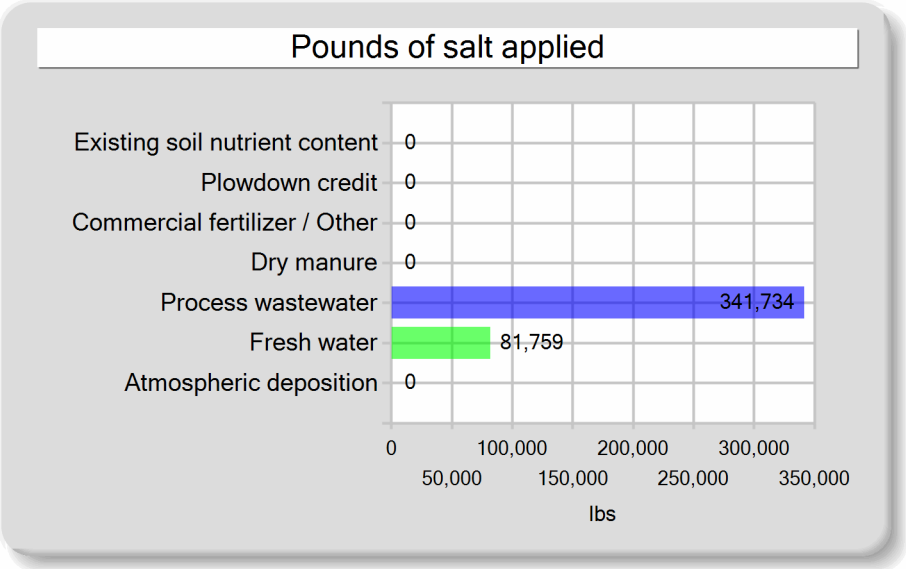


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Ken Van Foeken (Jun 21, 2024 10:28 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Ken VanFoeken	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
06/21/2024	
DATE	DATE

ATTACHMENT D

Manure Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: **Ken or Kurtis Van Foeken**

Name of Dairy Facility: **Ken Van Foeken Dairy #1**

Facility Address: **22338 Short Road, Hilmar, CA 95324**

Contact Person Name and Phone Number: **Kurtis Van Foeken (209) 658-1702**

Manure Hauler Information:

Name of Hauling Company/Person: **Silva & Sons, LLC**

Address of Hauling Company /Person: **23616 Williams Avenue, Hilmar, CA 95324**

Contact Person and Phone Number: **Frank Silva (209) 678-4027**

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) **Broker** (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Number and Street	City	Zip Code	Phone Number
Silva & Sons, LLC	23616 Williams Avenue	Hilmar	95324	(209) 678-4027

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Number and Street	City	Zip Code	Assessor's Parcel Number
23616 Williams Avenue	Hilmar	95324	

Dates Hauled: **5/19/23**

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: **1,200** Tons

Manure Solids Content (if amount reported in tons): **84.3%**

Method used to determine amount of manure: _____

Number of Truckloads x Average Truck Weight

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: **N/A** Gallons

Method used to determine volume of process wastewater: **N/A**

Written Agreement:

****NOT NECESSARY FOR SOLID MANURE MANIFEST****

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Ken Van Foeken (Jun 21, 2024 10:28 PDT) Date: 06/21/2024

Hauler's Signature: _____ Date: 5/19/2023

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Ken Van Foeken Dairy #1

Received: 11/28/2023 13:50
Reported: 11/30/2023 09:18

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23K1218-01	Well 1	Ag Water	T. Madruga		11/27/2023 12:30
23K1218-02	Well 2	Ag Water	T. Madruga		11/27/2023 12:38

Default Cooler Temperature on Receipt °C: 2.9
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Ken Van Foeken Dairy #1

Received: 11/28/2023 13:50
Reported: 11/30/2023 09:18

Sample Results

Sample: Well 1
23K1218-01 (Water)

Sampled: 11/27/2023 12:30
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.15**	mmhos/cm	0.01	1		11/29/23 14:08	SM 2510 B		BEK0810
Electrical Conductivity umhos	**1150**	umhos/cm	10.0	1		11/29/23 14:08	SM 2510 B		BEK0810
Ammonia (as N)	ND	mg/L	0.00	1		11/27/23 12:30	Field		BEK0821
Nitrate Nitrogen as NO3N	**54.8**	mg/L	0.1	1	10	11/28/23 21:32	EPA 300.0		BEK0809
Temperature	**25.0**	units	0.0	1		11/29/23 14:08	SM 4500-H+	H	BEK0810
pH	**7.7**	units	1.0	1		11/29/23 14:08	SM 4500-H+	H	BEK0810

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DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Ken Van Foeken Dairy #1

Received: 11/28/2023 13:50
Reported: 11/30/2023 09:18

Sample Results
(Continued)

Sample: Well 2
23K1218-02 (Water)

Sampled: 11/27/2023 12:38
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.33**	mmhos/cm	0.01	1		11/29/23 14:09	SM 2510 B		BEK0810
Electrical Conductivity umhos	**1330**	umhos/cm	10.0	1		11/29/23 14:09	SM 2510 B		BEK0810
Ammonia (as N)	ND	mg/L	0.00	1		11/27/23 12:38	Field		BEK0821
Nitrate Nitrogen as NO3N	**65.5**	mg/L	0.1	1	10	11/28/23 21:52	EPA 300.0		BEK0809
Temperature	**25.0**	units	0.0	1		11/29/23 14:09	SM 4500-H+	H	BEK0810
pH	**7.6**	units	1.0	1		11/29/23 14:09	SM 4500-H+	H	BEK0810

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Received: 11/28/2023 13:50
Reported: 11/30/2023 09:18

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0809										
Blank (BEK0809-BLK1)					Prepared & Analyzed: 11/28/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0809-BLK2)					Prepared & Analyzed: 11/28/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0809-BLK3)					Prepared: 11/28/2023 Analyzed: 11/29/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEK0809-BS1)					Prepared & Analyzed: 11/28/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		103	90-110		
LCS (BEK0809-BS2)					Prepared: 11/28/2023 Analyzed: 11/29/2023					
Nitrate Nitrogen as NO3N	5.5		0.1	mg/L	5.000		110	90-110		
Duplicate (BEK0809-DUP1)		**Source: 23K1216-02**			Prepared: 11/28/2023 Analyzed: 11/29/2023					
Nitrate Nitrogen as NO3N	4.1		0.1	mg/L		4.0			1.74	10
Duplicate (BEK0809-DUP2)		**Source: 23K1221-02**			Prepared: 11/28/2023 Analyzed: 11/29/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L		4.9			0.123	10
Matrix Spike (BEK0809-MS1)		**Source: 23K1216-02**			Prepared & Analyzed: 11/28/2023					
Nitrate Nitrogen as NO3N	9.2		0.1	mg/L	5.000	4.0	104	90-110		
Matrix Spike (BEK0809-MS2)		**Source: 23K1221-02**			Prepared: 11/28/2023 Analyzed: 11/29/2023					
Nitrate Nitrogen as NO3N	10.2		0.1	mg/L	5.000	4.9	107	90-110		
Reference (BEK0809-SRM1)					Prepared & Analyzed: 11/28/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEK0809-SRM2)					Prepared: 11/28/2023 Analyzed: 11/29/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEK0809-SRM3)					Prepared: 11/28/2023 Analyzed: 11/29/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

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Ranch: Ken Van Foeken Dairy #1

Received: 11/28/2023 13:50
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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0810										
Blank (BEK0810-BLK1)					Prepared: 11/28/2023	Analyzed: 11/29/2023				
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	6.2		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEK0810-BLK2)					Prepared: 11/28/2023	Analyzed: 11/29/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.8		1.0	units						
Blank (BEK0810-BLK3)					Prepared: 11/28/2023	Analyzed: 11/29/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
pH	7.8		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Duplicate (BEK0810-DUP1)		**Source: 23K1215-02**			Prepared: 11/28/2023	Analyzed: 11/29/2023				
Electrical Conductivity	1.83		0.01	mmhos/cm		1.85			0.766	10
Electrical Conductivity umhos	1830		10.0	umhos/cm		1850			0.766	10
pH	7.8		1.0	units		7.8			0.128	10
Duplicate (BEK0810-DUP2)		**Source: 23K1221-02**			Prepared: 11/28/2023	Analyzed: 11/29/2023				
Electrical Conductivity	0.46		0.01	mmhos/cm		0.46			0.810	10
Electrical Conductivity umhos	455		10.0	umhos/cm		459			0.810	10
pH	8.0		1.0	units		8.0			0.00	10
Reference (BEK0810-SRM1)					Prepared: 11/28/2023	Analyzed: 11/29/2023				
Electrical Conductivity	565			umhos/cm	538.0		105	90-110		
Reference (BEK0810-SRM2)					Prepared: 11/28/2023	Analyzed: 11/29/2023				
Electrical Conductivity	ND		0.01	mmhos/cm				90-110		
Reference (BEK0810-SRM3)					Prepared: 11/28/2023	Analyzed: 11/29/2023				
Electrical Conductivity	1070			umhos/cm	1000		107	90-110		
Reference (BEK0810-SRM4)					Prepared: 11/28/2023	Analyzed: 11/29/2023				
pH	6.6		1.0	units				97.5-102.5		
Reference (BEK0810-SRM5)					Prepared: 11/28/2023	Analyzed: 11/29/2023				
pH	6.5		1.0	units				97.5-102.5		
Reference (BEK0810-SRM6)					Prepared: 11/28/2023	Analyzed: 11/29/2023				

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0810 (Continued)										
Reference (BEK0810-SRM6)					Prepared: 11/28/2023	Analyzed: 11/29/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEK0810-SRM7)					Prepared: 11/28/2023	Analyzed: 11/29/2023				
pH	4.0			units	4.000		100	67021-101.3		

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11/28/23 13:50 23K1218

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

JG

No. Samples 2 No. of Bottles: 2

Purchase Order No | Bill To: | Acct # 19840 | Cons # 08

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Results Need By

Name: R & J Avila Inc
Address: 8119 Ellington Avenue
City: Hilmar **State:** CA **Zip:** 95324
Telephone: (209) 669-7919 **Fax:** (209) 634-4604
Cell/Email: 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com
COPY TO:
REQUESTED BY: Rui & John Avila
PROJECT Ken Van Foeken Dairy #1
ID CROP: N/A

[X] Copy of Chain [X] QA/QC Documents
Sampled By: T. Madruga

Other:
Analysis and Bottles Required: (Please indicate Analysis)
(X) DWW1: EC, pH, NO_3-N (NH4-N Field Test)
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DCW1: EC, NO_3-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	Well 1	11/27/2023	1230	2.9	ND
2	Well 2	11/27/2023	1238	2.8	ND
3					
4					
5					
6					
7					
8					
9					
10					

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	John Avila	R & J Avila, Inc.	11/27/2023 @ 1238	11/28/2023 @ 900
Second	JLF	Dellavalle Laboratory, Inc.	11/28/2023 @ 900	11-28-23 1:20
Third				
Fourth	JW	DL	11/28/23 13:50	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbit es. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all med rbitration,reasonable attorneys' fees of Dellavalle Laboratory.

IR Thermometer SN: 192603727
Correction Factor: 0°C
Calibration Due: 12/22/2023
Location: Laboratory

Invoicing Information:		*Shipping*	
Sampling hrs		$	In
Miles		$	Out
Consulting			
Amt Paid	Rec By	Check #	Date

Signature

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

11/28/23 13:50 23K1218

Shipping Information: Fed Ex □ UPS □ GSO □ Picked-Up ☒ Walk In □ DLI Sampler □
Ontrac □ PMS □ USPS □ Other □ _____

Shipping Container: Ice Chest ☒ None □ Box □

Refrigerant: Wet Ice: ☒ Blue Ice: □ None: □

Container(s) Received	1	2	3	4	5	6	7	8	9	10
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)										
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)	1	1								
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										